The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

The system trajectory of an asset is formalized as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

2.1 Corollary 1.1: Anthropological Incompleteness and Vacuum Collapse

When the external control vector m is stripped from the network, human judgment operates as a strictly closed formal system, F. By Gödel's First Incompleteness Theorem, a closed system cannot generate its own semantic validation ($F \nvdash (v_{\text{human}} = v)$). Syntactic legalism remains blind to absolute truth, causing a critical valuation $v_k \in V'$ to become structurally unobtainable:

$$\exists v_k [V'_{v_k} \wedge \neg h(u, v_k)] \tag{2}$$

Without the active insertion of the divine control input vector, the spatiotemporal execution plane collapses into an empty relational set ($[\,]$)—a mathematical vacuum. Christ operates as the external Meta-Language that breaks this internal syntax boundary limitation.

3 Sovereign Information-Theoretic Empowerment and Lot-Casting

We formalize the selection of Matthias (Acts 1:21–26) as an information-theoretic optimization executed by the Atemporal Controller (W_2), overriding the variance of closed human systems. Human agents lack the internal sensory technology to read hearts ($F \nvdash v$), reducing their subjective choices to uniform priors ($wpp = 0.50$).

3.1 The Divine Blahut-Arimoto Channel Capacity Matrix

The Atemporal Controller evaluates candidates by maximizing mutual information $I(X;Y)$ over the sensorimotor loop of the ecclesial environment, where X represents the action sequence space under environmental noise (e.g., rain or resource delays) and Y represents future state spaces. The capacity of the channel under sovereign optimization is governed by:

$$\text{Empowerment} = \max_{p(X)} \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x) p(y \mid x) \log \frac{p(y \mid x)}{\sum_{x'} p(x') p(y \mid x')} \tag{3}$$

Candidate A (Barsabbas) exhibits low empowerment ($\mathcal{E} = 0.3214$) because environmental perturbations collapse their action choices. Candidate B (Matthias) maintains high empowerment ($\mathcal{E} = 0.9418$) via scalable adaptive operational paths.

3.2 The Stochastic-Sovereign Lot Intercept and Inception

The casting of the lot functions as an explicit state-space intercept. By utilizing the `given` operator to remove human agency, the divine control vector m collapses the uniform distribution cleanly onto the maximum empowerment node. This culminates in an instantaneous Searlean Speech-Act Declaration, forcing a zero-uncertainty structural role mapping directly into the decision tree leaves:

$$\mathcal{D}_{\text{speech-act}}\Big(\langle \text{Church, Matthias, disciple} \rangle\Big) \implies \text{(Matthias counts as disciple in Church)} \tag{4}$$

This instantiates the historical institutional reality, allowing future structural queries for the twelve apostles to return Matthias with absolute semantic validation.